

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
```

Load Dataset

```
df=pd.read_csv("RSS - Original.csv")
df.head()
```

	X	Y	RSS - A	RSS - B	RSS - C
0	6	2	37	58	68
1	2	2	47	63	73
2	2	6	51	54	73
3	5	6	53	52	74
4	5	9	62	52	65

Next steps:

[Generate code with df](#)[New interactive sheet](#)

Check Dataset

```
print(df.shape)
```

```
(24, 5)
```

```
print(df.columns)
```

```
Index(['X', 'Y', 'RSS - A', 'RSS - B', 'RSS - C'], dtype='object')
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 24 entries, 0 to 23
Data columns (total 5 columns):
 #   Column      Non-Null Count  Dtype  
---  -
 0    X           24 non-null    int64  
 1    Y           24 non-null    int64  
 2   RSS - A     24 non-null    int64  
 3   RSS - B     24 non-null    int64  
 4   RSS - C     24 non-null    int64  
dtypes: int64(5)
memory usage: 1.1 KB
```

```
df.describe()
```

	X	Y	RSS - A	RSS - B	RSS - C
count	24.000000	24.000000	24.000000	24.000000	24.000000
mean	3.416667	18.541667	61.666667	61.375000	58.916667
std	2.041241	9.055285	9.581080	7.400426	9.357242
min	0.000000	2.000000	37.000000	39.000000	45.000000
25%	1.750000	11.250000	58.000000	58.000000	50.000000
50%	3.500000	20.000000	62.000000	62.500000	57.500000
75%	5.000000	25.250000	65.750000	66.000000	66.500000
max	6.000000	30.000000	82.000000	73.000000	74.000000

Handle Missing Value

```
df.isnull().sum()
```

```
      0
X      0
Y      0
RSS - A  0
RSS - B  0
RSS - C  0
```

```
dtype: int64
```

```
df=df.dropna()
```

```
df.drop_duplicates()
```

	X	Y	RSS - A	RSS - B	RSS - C	
0	6	2	37	58	68	
1	2	2	47	63	73	
2	2	6	51	54	73	
3	5	6	53	52	74	
4	5	9	62	52	65	
5	1	9	61	61	59	
6	5	12	55	39	74	
7	3	15	65	55	50	
8	6	16	54	58	60	
9	4	17	61	72	68	
10	4	20	63	58	53	
11	6	20	60	62	64	
12	0	20	82	69	57	
13	1	23	73	64	49	
14	4	23	62	66	45	
15	6	23	68	67	54	
16	6	25	63	62	49	
17	3	25	62	62	50	
18	1	26	73	66	66	
19	1	28	72	67	58	
20	3	28	59	66	50	
21	5	30	61	64	48	
22	3	30	62	63	51	
23	0	30	74	73	56	

Select Feature And Target

```
X=df[['X','Y']]
Y=df[['RSS - A','RSS - B','RSS - C']]
```

```
from sklearn.model_selection import train_test_split
X_train,X_test,Y_train,Y_test=train_test_split(X,Y,test_size=0.2,random_state=42)
```

```
from sklearn.linear_model import LinearRegression
model=LinearRegression()
model.fit(X_train,Y_train)
```

▼ LinearRegression ⓘ ?

LinearRegression()

```
Y_pred=model.predict(X_test)
```

```
from sklearn.metrics import mean_absolute_error,mean_squared_error,r2_score
import numpy as np
print(mean_absolute_error(Y_test,Y_pred))
print(mean_squared_error(Y_test,Y_pred))
print(np.sqrt(mean_squared_error(Y_test,Y_pred)))
print(r2_score(Y_test,Y_pred))
```

```
4.871763234024626
38.173134009773726
6.178441066302545
-0.48447434144451784
```

```
new_location=[[4,3]]
pred=model.predict(new_location)
print(pred)
```

```
[[52.85955747 52.27004527 71.63198431]]
/usr/local/lib/python3.13/dist-packages/sklearn/utils/validation.py:2739: UserWarning:
  warnings.warn(
```

```
result=pd.DataFrame({
    'Actual RSS - A':Y_test['RSS - A'].values,
    'pred RSS - A':Y_pred[:,0],
    'Actual RSS - B':Y_test['RSS - B'].values,
    'pred RSS - B':Y_pred[:,1],
    'Actual RSS - C':Y_test['RSS - C'].values,
    'pred RSS - C':Y_pred[:,2]})
```